

DBMS – ORDER AND ORDER_ITEM MANAGEMENT

1. DATABASE DESIGN

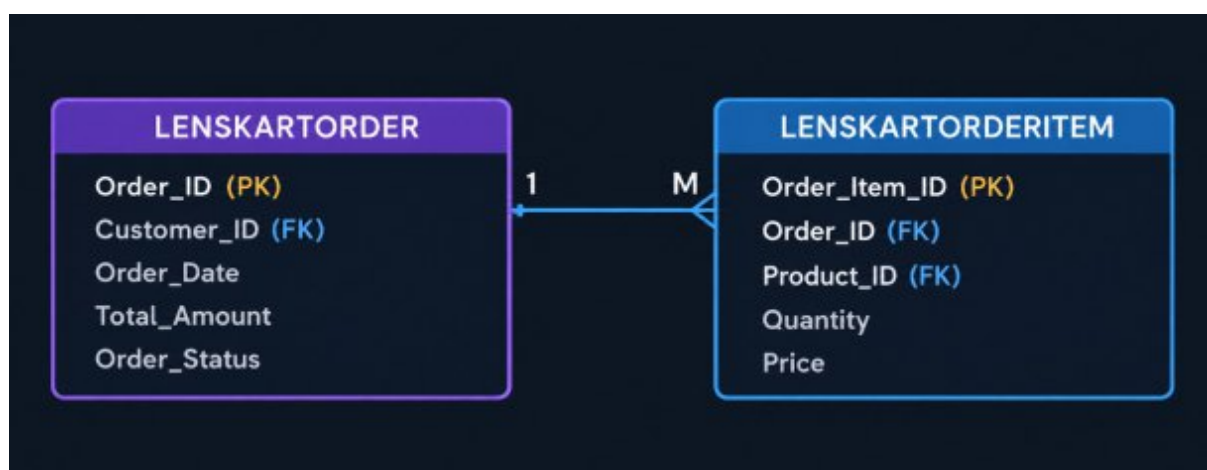
1.1 LENSKART ORDER TABLE

Attribute	Description
Order_ID	Unique order ID
Customer_ID	Customer ID
Order_Date	Date of order
Total_Amount	Total amount
Order_Status	Order status

1.2 LENSKART ORDER_ITEM TABLE

Attribute	Description
Order_Item_ID	Unique order item ID
Order_ID	Order ID
Product_ID	Product ID
Quantity	Quantity ordered
Price	Product price

2.RELATIONSHIP



3.TABLE CREATION

3.1 ORDER TABLE

```
SQL> CREATE TABLE LenskartOrder (  
  2     Order_ID NUMBER PRIMARY KEY,  
  3     Customer_ID NUMBER,  
  4     Order_Date DATE NOT NULL,  
  5     Total_Amount NUMBER(10,2) NOT NULL,  
  6     Order_Status VARCHAR2(30) NOT NULL,  
  7     FOREIGN KEY (Customer_ID) REFERENCES LenskartCustomer(Customer_ID)  
  8 );
```

Table created.

3.2 ORDER_ITEM TABLE

```
SQL> CREATE TABLE LenskartOrderItem (  
  2     Order_Item_ID NUMBER PRIMARY KEY,  
  3     Order_ID NUMBER,  
  4     Product_ID NUMBER,  
  5     Quantity NUMBER NOT NULL,  
  6     Price NUMBER(10,2) NOT NULL,  
  7     FOREIGN KEY (Order_ID) REFERENCES LenskartOrder(Order_ID),  
  8     FOREIGN KEY (Product_ID) REFERENCES LenskartProduct(Product_ID)  
  9 );
```

Table created.

4.ORDER TABLE DATA INSERTION

```
SQL> INSERT INTO LenskartOrder  
  2 VALUES (1, 1, TO_DATE('18-09-2026','DD-MM-YYYY'), 1200, 'Confirmed');
```

1 row created.

SQL>

```
SQL> INSERT INTO LenskartOrder  
  2 VALUES (2, 2, TO_DATE('19-09-2026','DD-MM-YYYY'), 1800, 'Shipped');
```

1 row created.

SQL>

```
SQL> INSERT INTO LenskartOrder  
  2 VALUES (3, 3, TO_DATE('20-09-2026','DD-MM-YYYY'), 1500, 'Delivered');
```

1 row created.

SQL>

```
SQL> INSERT INTO LenskartOrder  
  2 VALUES (4, 4, TO_DATE('21-09-2026','DD-MM-YYYY'), 2000, 'Confirmed');
```

1 row created.

SQL>

```
SQL> INSERT INTO LenskartOrder  
  2 VALUES (5, 5, TO_DATE('22-09-2026','DD-MM-YYYY'), 900, 'Processing');
```

1 row created.

4.1 DISPLAY ORDER TABLE DATA

```
SQL> SELECT * FROM LenskartOrder;
```

ORDER_ID	CUSTOMER_ID	ORDER_DAT	TOTAL_AMOUNT	ORDER_STATUS
1	1	18-SEP-26	1200	Confirmed
2	2	19-SEP-26	1800	Shipped
3	3	20-SEP-26	1500	Delivered
4	4	21-SEP-26	2000	Confirmed
5	5	22-SEP-26	900	Processing

5.ORDER_ITEM TABLE DATA INSERTION

```
SQL> INSERT INTO LenskartOrderItem
2 VALUES (1, 1, 1, 1, 1200);
```

```
1 row created.
```

```
SQL>
SQL> INSERT INTO LenskartOrderItem
2 VALUES (2, 2, 2, 1, 1800);
```

```
1 row created.
```

```
SQL>
SQL> INSERT INTO LenskartOrderItem
2 VALUES (3, 3, 3, 1, 1500);
```

```
1 row created.
```

```
SQL>
SQL> INSERT INTO LenskartOrderItem
2 VALUES (4, 4, 4, 1, 2000);
```

```
1 row created.
```

```
SQL>
SQL> INSERT INTO LenskartOrderItem
2 VALUES (5, 5, 5, 1, 900);
```

```
1 row created.
```

5.1 DISPLAY ORDER_ITEM TABLE DATA

```
SQL> SELECT * FROM LenskartOrderItem;
```

ORDER_ITEM_ID	ORDER_ID	PRODUCT_ID	QUANTITY	PRICE
1	1	1	1	1200
2	2	2	1	1800
3	3	3	1	1500
4	4	4	1	2000
5	5	5	1	900

```
SQL> |
```

6. ORDER MODIFICATION OPERATION

6.1 MODIFY ORDER AMOUNT AND STATUS

```
SQL> UPDATE LenskartOrder
  2  SET Total_Amount = 2400,
  3      Order_Status = 'Confirmed'
  4  WHERE Order_ID = 1;

1 row updated.
```

```
SQL> UPDATE LenskartOrder
  2  SET Total_Amount = 3600,
  3      Order_Status = 'Shipped'
  4  WHERE Order_ID = 2;

1 row updated.
```

```
SQL> SELECT * FROM LenskartOrder;

ORDER_ID  CUSTOMER_ID  ORDER_DAT  TOTAL_AMOUNT  ORDER_STATUS
-----
      1           1  18-SEP-26      2400  Confirmed
      2           2  19-SEP-26      3600  Shipped
      3           3  20-SEP-26      1500  Delivered
      4           4  21-SEP-26      2000  Confirmed
      5           5  22-SEP-26       900  Processing
```

7. ORDER_ITEM MODIFICATION

7.1 MODIFY ORDER QUALITY AND PRICE

```
SQL> UPDATE LenskartOrderItem
  2  SET Quantity = 2,
  3      Price = 1200
  4  WHERE Order_Item_ID = 1;

1 row updated.
```

```
SQL> UPDATE LenskartOrderItem
  2  SET Quantity = 2,
  3      Price = 1800
  4  WHERE Order_Item_ID = 2;

1 row updated.
```

```
SQL> SELECT * FROM LenskartOrderItem;
```

ORDER_ITEM_ID	ORDER_ID	PRODUCT_ID	QUANTITY	PRICE
1	1	1	2	1200
2	2	2	2	1800
3	3	3	1	1500
4	4	4	1	2000
5	5	5	1	900

8. CUSTOMER ORDER HISTORY REPORT

```
SQL> SELECT
  2     c.Customer_ID AS Customer_ID,
  3     c.Customer_Name AS Customer_Name,
  4     o.Order_ID,
  5     o.Order_Date,
  6     oi.Product_ID,
  7     p.Product_Name,
  8     oi.Quantity,
  9     oi.Price,
 10     (oi.Quantity * oi.Price) AS Total_Amount
 11 FROM LenskartCustomer c
 12 JOIN LenskartOrder o
 13 ON c.Customer_ID = o.Customer_ID
 14 JOIN LenskartOrderItem oi
 15 ON o.Order_ID = oi.Order_ID
 16 JOIN LenskartProduct p
 17 ON oi.Product_ID = p.Product_ID
 18 ORDER BY c.Customer_ID, o.Order_Date;
```

```
-----
CUSTOMER_NAME
-----
ORDER_ID ORDER_DAT PRODUCT_ID
-----
PRODUCT_NAME
-----
QUANTITY    PRICE  TOTAL_AMOUNT
-----
1
Arun Kumar
1 18-SEP-26      1
```

```
CUSTOMER_ID
-----
CUSTOMER_NAME
-----
ORDER_ID ORDER_DAT PRODUCT_ID
-----
PRODUCT_NAME
-----
QUANTITY    PRICE  TOTAL_AMOUNT
-----
Classic Black Frame
2      1200      2400
```

```
CUSTOMER_ID
-----
CUSTOMER_NAME
-----
ORDER_ID ORDER_DAT PRODUCT_ID
-----
PRODUCT_NAME
-----
QUANTITY    PRICE  TOTAL_AMOUNT
-----
2
Priya Sharma
2 19-SEP-26      2
```

```
CUSTOMER_ID
-----
CUSTOMER_NAME
-----
ORDER_ID ORDER_DAT PRODUCT_ID
-----
PRODUCT_NAME
-----
QUANTITY    PRICE TOTAL_AMOUNT
-----
Blue Light Glasses
      2      1800      3600
```

```
CUSTOMER_ID
-----
CUSTOMER_NAME
-----
ORDER_ID ORDER_DAT PRODUCT_ID
-----
PRODUCT_NAME
-----
QUANTITY    PRICE TOTAL_AMOUNT
-----
      3
Rahul Kumar
      3 20-SEP-26      3
```

```
CUSTOMER_ID
-----
CUSTOMER_NAME
-----
ORDER_ID ORDER_DAT PRODUCT_ID
-----
PRODUCT_NAME
-----
QUANTITY    PRICE TOTAL_AMOUNT
-----
Round Metal Frame
      1      1500      1500
```

```
CUSTOMER_ID
-----
CUSTOMER_NAME
-----
ORDER_ID ORDER_DAT PRODUCT_ID
-----
PRODUCT_NAME
-----
QUANTITY    PRICE TOTAL_AMOUNT
-----
4
Divya Raj
4 21-SEP-26      4
```

```
CUSTOMER_ID
-----
CUSTOMER_NAME
-----
ORDER_ID ORDER_DAT PRODUCT_ID
-----
PRODUCT_NAME
-----
QUANTITY    PRICE TOTAL_AMOUNT
-----
Classic Black Sunglasses
1      2000      2000
```

```
CUSTOMER_ID
-----
CUSTOMER_NAME
-----
ORDER_ID ORDER_DAT PRODUCT_ID
-----
PRODUCT_NAME
-----
QUANTITY    PRICE TOTAL_AMOUNT
-----
5
Karthik S
5 22-SEP-26      5
```

```
CUSTOMER_ID
-----
CUSTOMER_NAME
-----
ORDER_ID ORDER_DAT PRODUCT_ID
-----
PRODUCT_NAME
-----
QUANTITY    PRICE TOTAL_AMOUNT
-----
Daily Contact Lenses
1      900      900
```

9. CUSTOMER WISE ORDER SUMMARY

```
SQL> SELECT
  2     c.Customer_ID AS Customer_ID,
  3     c.Customer_Name AS Customer_Name,
  4     COUNT(o.Order_ID) AS Total_Orders,
  5     SUM(o.Total_Amount) AS Total_Amount
  6 FROM LenskartCustomer c
  7 JOIN LenskartOrder o
  8 ON c.Customer_ID = o.Customer_ID
  9 GROUP BY c.Customer_ID, c.Customer_Name
 10 ORDER BY c.Customer_ID;
```

CUSTOMER_ID

CUSTOMER_NAME

TOTAL_ORDERS TOTAL_AMOUNT

1
Arun Kumar
1 2400

2
Priya Sharma
1 3600

CUSTOMER_ID

CUSTOMER_NAME

TOTAL_ORDERS TOTAL_AMOUNT

3
Rahul Kumar
1 1500

4
Divya Raj

CUSTOMER_ID

CUSTOMER_NAME

TOTAL_ORDERS TOTAL_AMOUNT

1 2000

5
Karthik S
1 900